

# Fibonacci Polynomials & *Lights Out*

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## Abstract

Consider a game played on a simple graph  $G = (V, E)$  where each vertex consists of clickable light. A move is clicking any  $v \in V$ , which toggles the on/off state of  $v$  and its neighbors. Given an initial configuration, one wins the game by finding a sequence of moves that turns off all of the lights. This game was commercially available from Tiger Electronics as *Lights Out* for  $G$  a  $5 \times 5$  grid.

Let  $\Phi_G : P(V) \rightarrow P(V)$  be the function which maps the vertices clicked to the vertices that change state. For  $G$  and  $n \times n$  grid, let  $d(n) = \dim(\ker \Phi_G)$ . We show several that many values of  $d(n)$  never occur. We show more general formulas for the value of  $d(n)$ , including a complete characterization when  $n = 2^a p^b - 1$  for  $a, b \geq 0$  and  $p$  not a Wieferich prime.

## 1 Fibonacci Polynomials

**Definition 1.1** (Fibonacci Polynomial). *Let  $F_n(x)$  be the polynomial in the ring  $\mathbb{F}_2[x]$  defined recursively by*

$$F_n(x) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ xF_{n-1}(x) + F_{n-2}(x) & \text{otherwise} \end{cases}.$$

These polynomials are called "Fibonacci" because when defined over the real numbers,  $F_n(1)$  is the  $n$ th Fibonacci number.

In  $\mathbb{F}_2[x]$ , the coefficients of  $F_n(x)$  are either 1 or 0, so we can visualize them by coloring squares black or white to represent the coefficients. For example,

$$F_6(x) = 1x^5 + 0x^4 + 0x^3 + 0x^2 + 1x + 0 = \blacksquare \square \square \square \blacksquare \square.$$

Plotting  $F_1(x)$  through  $F_{128}(x)$  in the same way, aligning terms of the same degree, we see a Sierpinski triangle.

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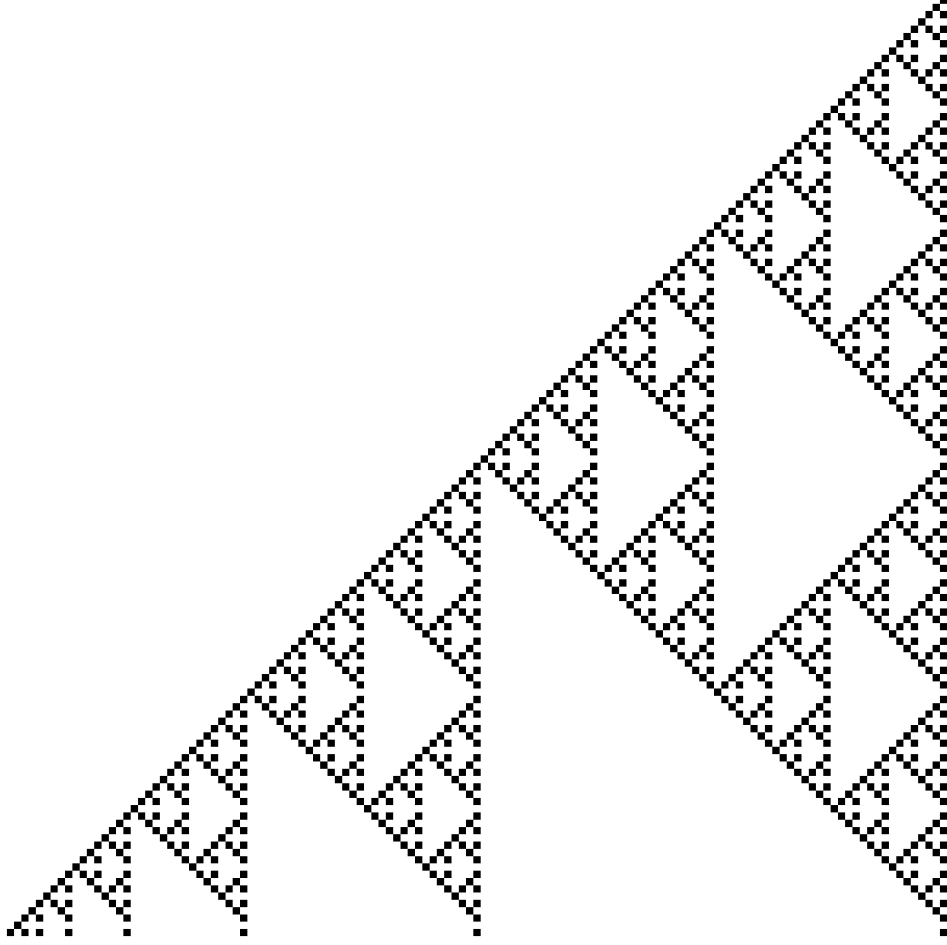


Figure 1:  $F_1(x)$  through  $F_{128}(x)$ , right-aligned.

Using a well-known connection between the Sierpinski triangle and the parity of binomial coefficients, we may see that

$$F_n(x) = \sum_{i=0}^{n-1} \binom{n+i}{2i+1} x^i \pmod{2}.$$

The above construction and formula was also shown by Sutner [10]. Sutner also showed the following relationship between square *Lights Out* grids and Fibonacci Polynomials.

**Lemma 1.1** (Sutner). *Let  $G = (V, E)$  be an  $n \times m$  grid graph. Let  $\Phi_G : P(V) \rightarrow P(V)$  be the function that maps which lights are clicked to which lights change state. Then*

$$\dim(\ker \Phi_G) = \deg(\gcd(F_{n+1}(x), F_{m+1}(x+1))) = \deg(\gcd(F_{n+1}(x+1), F_{m+1}(x))).$$

*Proof.* See [10] theorem 5.1. ■

Sutner and others also shows several divisibility rules for these polynomials

**Lemma 1.2** (Hunziker, Machiavelo, and Park). *A polynomial  $\tau(x)$  in  $\mathbb{F}_2[x]$  divides both  $F_n(x)$  and  $F_m(x)$  if and only if it divides  $F_{\gcd(m,n)}(x)$ . In particular,*

$$\gcd(F_m(x), F_n(x)) = F_{\gcd(m,n)}(x).$$

*Proof.* See [5] proposition 2.2, [10] proposition 2.1, and [6]. ■

**Corollary 1.2.1.** *The following are true:*

- (i)  $x$  divides  $F_n(x)$  if and only if  $n \equiv 0 \pmod{2}$ .
- (ii)  $x + 1$  divides  $F_n(x + 1)$  if and only if  $n \equiv 0 \pmod{2}$ .
- (iii)  $x + 1$  divides  $F_n(x)$  if and only if  $n \equiv 0 \pmod{3}$ .
- (iv)  $x$  divides  $F_n(x + 1)$  if and only if  $n \equiv 0 \pmod{3}$ .

*Proof.* Notice,

- (i) We find that  $f_2(x) = x$  is the smallest Fibonacci polynomial divisible by  $x$ , so we apply lemma 1.2 to get the desired result.
  - (ii) Follows from (i) by substituting  $x + 1$  for  $x$ .
  - (iii) We find that  $f_3(x) = (x + 1)^2$  is the smallest Fibonacci polynomial divisible by  $x + 1$ , so we apply lemma 1.2 to get the desired result.
  - (iv) Follows from (iii) by substituting  $x + 1$  for  $x$ .
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**Lemma 1.3** (Hunziker, Machiavelo, and Park). *If  $n$  is odd, then  $F_n(x)$  is the square of a square-free polynomial.*

*Proof.* See [5] proposition 2.11 and [10] Theorem 2.1 claim 2. ■

**Corollary 1.3.1.** *If  $n$  is odd, then every root of  $F_n(x)$  has multiplicity 2.*

*Proof.* Let  $n$  be odd and  $\alpha$  a root of  $F_n(x)$ . By 1.3, there exists  $\tau_n \in \mathbb{F}_2[x]$  such that

$$F_n(x) = \tau_n(x)^2,$$

and  $\tau(x)$  is square-free. Since  $\alpha$  is a root of  $F_n(x)$ , it must also be a root of  $\tau_n(x)$  and thus has multiplicity 2. ■

We can now prove some of our own facts about these polynomials and their GCDs, particularly on square grids.

**Lemma 1.4.** *Let  $d(n) = \dim(\ker \Phi_G)$  for  $G$  an  $n \times n$  grid graph. Then  $d(n)$  is always even.*

*Proof.* Let

$$G_n(x) = \gcd(F_n(x), F_n(x + 1)).$$

By lemma 1.1, we know that  $d(n) = \deg G_{n+1}(x)$  and  $G_n(x) = G_n(x + 1)$ .

$y = x(x + 1)$  is also unchanged by this  $x \mapsto x + 1$  translation. It is well-known that every polynomial  $P(x)$  in  $\mathbb{F}_2[x]$  can be uniquely written as

$$P(x) = A(y) + xB(y) \quad A, B \in \mathbb{F}_2[y].$$

We can then determine exactly when  $P$  is also unchanged by translation.

$$\begin{aligned} P(x+1) &= P(x) \\ A(y) + (x+1)B(y) &= A(y) + xB(y) \\ B(y) &= 0. \end{aligned}$$

In other words,  $P(x)$  is unchanged by translation when  $P$  is a polynomial in  $x^2 + x$ .

Since  $G_n$  is also unchanged by translation, there exists  $H_n \in \mathbb{F}_2[x]$  such that

$$G_n(x) = H_n(x^2 + x).$$

So,

$$d(n) = \deg G_{n+1}(x) = 2 \deg H_{n+1}(x)$$

and thus  $d(n)$  is even. ■

**Corollary 1.4.1.** *If  $n$  is even, then  $d(n)$  is divisible by 4.*

*Proof.* If  $n$  is even, then  $n+1$  is odd. Then by 1.3,  $F_{n+1}(x)$  and  $F_{n+1}(x+1)$  are squares of square-free polynomials  $\tau(x)$  and  $\tau(x+1)$ .

$$\begin{aligned} G_{n+1}(x) &= \gcd(F_{n+1}(x), F_{n+1}(x+1)) \\ &= \gcd(\tau_{n+1}(x)^2, \tau_{n+1}(x+1)^2) \\ &= \gcd(\tau_{n+1}(x), \tau_{n+1}(x+1))^2 \\ \sqrt{G_{n+1}(x)} &= \gcd(\tau_{n+1}(x), \tau_{n+1}(x+1)) \end{aligned}$$

We can see from the right-hand side of the equation that this function is also unaffected by translation. So, by lemma 1.4, there exists  $H_{n+1} \in \mathbb{F}_2[x]$  such that

$$\begin{aligned} \sqrt{G_{n+1}(x)} &= H_{n+1}(x^2 + x) \\ G_{n+1}(x) &= H_{n+1}(x^2 + x)^2. \end{aligned}$$

So,

$$d(n) = \deg G_{n+1}(x) = 4 \deg H_{n+1}(x),$$

as desired. ■

Note that the implication does not hold in the other direction:  $d(n)$  also be divisible by 4 when  $n$  is odd. For example,  $d(9) = 8$ .

**Lemma 1.5.** *Let  $x = t + t^{-1}$ . Then for  $t \neq 0$ ,*

$$xF_n(x) = t^n + t^{-n}.$$

*Proof.* Define  $G_n(t) = t^{n+1} + t^{-(n+1)}$ . Then

$$\begin{aligned} xG_{n-1}(t) + G_{n-2}(t) &= (t + t^{-1})(t^n + t^{-n}) + (t^{n-1} + t^{-(n-1)}) \\ &= t^{n+1} + t^{-(n-1)} + t^{n-1} + t^{-(n+1)} + t^{n-1} + t^{-(n-1)} \\ &= t^{n+1} + t^{-(n+1)} \\ &= G_n. \end{aligned}$$

Thus we see that  $G_n(t)$  satisfies the same recurrence relation as  $F_n(x)$ .  $xF_n(x)$  also satisfies the same recurrence relation.

$$\begin{aligned} xF_n(x) &= x(xF_{n-1}(x) + F_{n-2}(x)) \\ &= x(xF_{n-1}(x)) + xF_{n-2}(x). \end{aligned}$$

Now we compute  $xF_1(x)$  and  $xF_2(x)$  in terms of  $t$  and  $G$ :

$$\begin{aligned} xF_1(x) &= x \\ &= t + t^{-1} \\ &= G_0(t), \end{aligned}$$

and

$$\begin{aligned} xF_2(x) &= x^2 \\ &= (t + t^{-1})^2 \\ &= t^2 + t^{-2} \\ &= G_1(t). \end{aligned}$$

Since the first two terms match and they satisfy the same recurrence relation then we've shown by induction that for  $t \neq 0$ ,

$$xF_n(x) = G_{n-1}(t) = t^n + t^{-n}.$$

■

This was also proven in [5], who also prove the following identities.

**Lemma 1.6** (Hunziker, Machiavelo, and Park). *Let  $n = 2^k b$  where  $b, k \in \mathbb{N}$ . Then*

$$F_n(x) = x^{2^k - 1} F_b(x)^{2^k}.$$

*Proof.* See [5] lemma 2.6. ■

**Lemma 1.7.** *The distinct roots of  $F_n(x)$  are  $\{\zeta + \zeta^{-1} : \zeta^n = 1, \zeta \neq 1\}$ , plus 0 when  $n$  is even.*

*Proof.* Notice that  $F_n(0) = F_{n-2}(0)$ , so

$$F_n(0) = \begin{cases} 1 & n \text{ odd} \\ 0 & n \text{ even} \end{cases}.$$

Thus 0 is a root of  $F_n(x)$  whenever  $n$  is even.

Let  $\alpha$  be a non-zero root of  $F_n$ . Choose  $\zeta$  satisfying

$$\zeta^2 + \alpha\zeta + 1 = 0.$$

Notice that  $\zeta$  cannot be 0. Dividing by  $\zeta$ ,

$$\begin{aligned} \zeta + \alpha + \zeta^{-1} &= 0 \\ \alpha &= \zeta + \zeta^{-1}. \end{aligned}$$

By, Lemma 1.5,

$$\begin{aligned} 0 &= \alpha F_n(\alpha) \\ &= \zeta^n + \zeta^{-n} \end{aligned}$$

Since a field has no nonzero nilpotents, we may multiply by  $\zeta^n$ .

$$\begin{aligned} 0 &= \zeta^{2n} + 1 \\ &= (\zeta^n + 1)^2, \end{aligned}$$

and

$$\begin{aligned} 0 &= \zeta^n + 1 \\ \zeta^n &= 1. \end{aligned}$$

Thus the order of  $\zeta$  divides  $n$ . Notice too that  $\zeta \neq 1$ ; otherwise,  $\alpha = 1 + 1^{-1} = 0$ , which we already showed as not possible.

Conversely, if  $\zeta^n = 1$  and  $\zeta \neq 1$ , then let  $\alpha = \zeta + \zeta^{-1}$ . Then we see  $\alpha \neq 0$  and

$$\alpha F_n(\alpha) = \zeta^n + \zeta^{-n} = 1 + 1^{-1} = 0.$$

Thus,  $\alpha$  is a root of  $F_n$ . ■

## 2 Even Parity Covers

$\ker \Phi_G$  may also be understood via parity covers and dominating sets.

**Definition 2.1.** *An even dominating set of a graph  $G = (V, E)$  is a subset  $V' \subseteq V$  such that every vertex contains an even number of vertices from  $V'$  in its neighborhood.*

Odd dominating sets can be analogously defined. Sutner specifically studied odd dominating sets as the solution to *Lights Out* when all lights are initially on. He showed that such an odd dominating set always exists for any finite graph [9].

**Lemma 2.1.**  *$V' \in \ker \Phi_G$  if and only if  $V'$  is an even dominating set of  $G$ .*

*Proof.* Let  $V' \in \ker \Phi_G$ . Let  $N(v)$  be the neighborhood of vertex  $v$ :  $v$  and its immediate neighbors. Then for every  $v \in V$ ,

$$|N(v) \cap V'| \text{ is even.}$$

Thus,  $V'$  is an even dominating set of  $G$ . We can apply the same steps in reverse to show the converse. ■

**Lemma 2.2.**  *$\Phi_G$  is linear with respect to symmetric difference. That is, for all  $V_1, V_2 \subseteq V$ ,*

$$\Phi_G(V_1 \triangle V_2) = \Phi_G(V_1) \triangle \Phi_G(V_2).$$

*Proof.*

$$\begin{aligned} \Phi_G(V_1 \triangle V_2) &= \{v \in V : |N(v) \cap (V_1 \triangle V_2)| \text{ is odd}\} \\ &= (\{v \in V : |N(v) \cap V_1| \text{ is odd}\} \cup \{v \in V : |N(v) \cap V_2| \text{ is odd}\}) \setminus \{v \in V : |N(v) \cap (V_1 \cap V_2)| \text{ is odd}\} \\ &= \{v \in V : |N(v) \cap V_1| \text{ is odd}\} \triangle \{v \in V : |N(v) \cap V_2| \text{ is odd}\} \\ &= \Phi_G(V_1) \triangle \Phi_G(V_2) \end{aligned}$$
■

**Corollary 2.2.1.** *If  $V_1$  and  $V_2$  have the same image under  $\Phi_G$  if and only if  $V_1 \triangle V_2 \in \ker \Phi_G$ .*

*Proof.* Assume that  $V_1$  and  $V_2$  have the same image under  $\Phi_G$ . Then

$$\begin{aligned} \Phi_G(V_1 \triangle V_2) &= \Phi_G(V_1) \triangle \Phi_G(V_2) \\ &= \Phi_G(V_1) \triangle \Phi_G(V_1) \\ &= \emptyset. \end{aligned}$$

Thus,  $V_1 \triangle V_2 \in \ker \Phi_G$ . Now assume  $V_1 \triangle V_2 \in \ker \Phi_G$ . Then

$$\begin{aligned}\Phi_G(V_1) &= \Phi_G(V_1) \triangle \emptyset \\ &= \Phi_G(V_1) \triangle \Phi_G(V_1 \triangle V_2) \\ &= \Phi_G(V_1) \triangle \Phi_G(V_1) \triangle \Phi_G(V_2) \\ &= \Phi_G(V_2).\end{aligned}$$

Thus  $V_1$  and  $V_2$  have the same image under  $\Phi_G$ . ■

We may now use our geometric understanding of  $\ker \Phi_G$  as even parity covers to construct a lower bound on  $d(n)$ .

**Theorem 2.1.** *For all  $n, k \in \mathbb{N}$ ,*

$$d(nk - 1) \geq d(n - 1).$$

*Proof.* Let  $G_L$  be an  $nk - 1 \times nk - 1$  grid and  $G_\ell$  be an  $n - 1 \times n - 1$  grid. Since  $nk - 1 = k(n - 1) + (k - 1)$ , an  $G$  may be broken into a  $k \times k$  grid of  $n - 1 \times n - 1$  grids, each separated by a horizontal or vertical strip of size 1.

Let  $Q$  be an even dominating set of  $G_\ell$ . Let  $R$  be  $Q$  tiled  $k$  times horizontally and vertically, with size 1 separators between each tile, onto  $G_L$  where each tile is the reflection across its neighboring separators, as shown below.

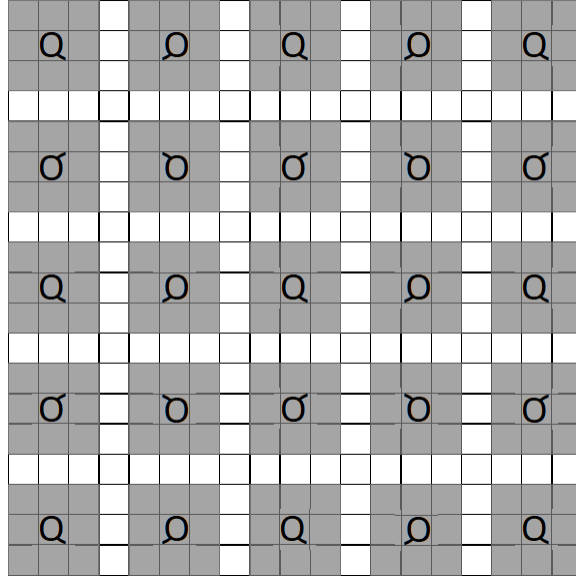


Figure 2: Pattern  $Q$  tiled 5 times.

Consider a vertex  $v$  in  $G_L$ . There are 3 cases.

1.  $v$  is within a tile. Then  $|N(v) \cap R|$  is even because  $Q$  being is an even parity cover.
2.  $v$  is on a separator between two tiles. Then only two elements in  $N(v)$ , the vertices within a tile, are also in  $R$ . Because neighboring tiles are reflections of each other across separators, these two vertices are either both in  $R$  or both not in  $R$ . In either case,  $|N(v) \cap R|$  is even.
3.  $v$  is in the intersection of two separators. Then all of  $N(v)$  is in a separator and thus not in  $R$ , so  $|N(v) \cap R| = 0$ , which is even.

Since for every vertex  $v$  of  $G_L$ ,  $|N(v) \cap R|$  is even, so  $R$  is an even dominating set of  $G_L$ . Thus, for every even dominating set of  $G_\ell$ , there exists an even dominating set of  $G_L$  obtained through tiling. This tiling function is also injective because we can recover  $Q$  by restricting  $R$  to just the top left tile. Thus,

$$d(nk - 1) \geq d(n - 1),$$

as desired. ■

Sutner shows a similar result [8]. There is also a simple proof in terms of Fibonacci polynomials that also provides a divisibility relationship.

*Proof.* Since  $n$  divides  $nk$ , lemma 1.2 tells us  $F_n(x)$  divides  $F_{nk}(x)$ . So,

$$\gcd(F_n(x), F_n(x+1)) \mid \gcd(F_{nk}(x), F_{nk}(x+1)).$$

So we may write

$$\gcd(F_{nk}(x), F_{nk}(x+1)) = \gcd(F_n(x), F_n(x+1))Q(x),$$

for some  $Q(x) \in \mathbb{F}_2[x]$ . Taking the degrees, we obtain  $d(nk - 1) = d(n - 1) + \deg Q(x)$ , and so the desired inequality follows. ■

Note that although the smaller GCD polynomial divides the larger, it's not the case that  $d(n - 1)$  divides  $d(nk - 1)$ . For example, if  $n = 5$  and  $k = 6$ , then  $d(n - 1) = d(4) = 4$ , which does not divide  $d(nk - 1) = d(29) = 10$ .

### 3 Number Theory

We may also understand  $d(n)$  and the Fibonacci polynomials with some results from number theory. Many of the results in this section were first discussed by the author in [2] and [1]. However, this writing restates several proofs for clarity and to correct some small mistakes in those earlier proofs.

**Lemma 3.1** (Ore). *Let  $a, b, c$ , and  $d$  be integers. Let  $(a, b)$  denote  $\gcd(a, b)$ . Then*

$$(ab, cd) = (a, c)(b, d) \left( \frac{a}{(a, c)}, \frac{d}{(b, d)} \right) \left( \frac{c}{(a, c)}, \frac{b}{(b, d)} \right).$$

*Proof.* See [11]. ■

Although Ore's result deals specifically with integers, both the integers and  $\mathbb{F}_2[x]$  are Euclidean domains, so the result will still hold for our purposes.

**Theorem 3.1.** *For all  $n \in \mathbb{N}$ ,*

$$d(2n + 1) = 2d(n) + \delta_n, \quad \delta_n = \begin{cases} 2 & 3 \mid n + 1 \\ 0 & \text{otherwise} \end{cases}.$$



*Proof.* Let  $(a, b)$  denote  $\gcd(a, b)$ . Applying lemmas 1.6 and 3.1,

$$\begin{aligned}
d(2n+1) &= \deg(F_{2n+2}(x), F_{2n+2}(x+1)) \\
&= \deg(xF_{n+1}(x)^2, (x+1)F_{n+1}(x+1)^2) \\
&= \deg(x, x+1) (F_{n+1}(x)^2, F_{n+1}(x+1)^2) \left( \frac{x+1}{(x, x+1)}, \frac{F_{n+1}(x)^2}{(F_{n+1}(x)^2, F_{n+1}(x+1)^2)} \right) \left( \frac{x}{(x, x+1)}, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x)^2, F_{n+1}(x+1)^2)} \right) \\
&= \deg(F_{n+1}(x), F_{n+1}(x+1))^2 \left( x+1, \frac{F_{n+1}(x)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right) \left( x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right) \\
&= 2d(n) + 2 \deg \left( x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right)
\end{aligned}$$

Thus, we have that

$$\delta_n = 2 \deg \left( x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right).$$

So,  $\delta_n$  is 2 exactly when  $F_{n+1}(x)$  can be divided without remainder by  $x+1$  more times than  $F_{n+1}(x+1)$ . When  $n+1$  is not divisible by 3, corollary 1.2.1 tells us that  $F_{n+1}(x)$  is not divisible by  $x+1$ , so  $\delta_n = 0$ . When  $n+1$  is divisible by 3, let  $n+1 = b2^k$  for  $b, k \geq 0$  and  $b$  odd. Then  $b$  must be divisible by 3. Applying Lemma 1.6,

$$F_{n+1}(x) = x^{2^k-1} F_b(x)^{2^k} \quad F_{n+1}(x+1) = (x+1)^{2^k-1} F_b(x+1)^{2^k}.$$

Since  $b$  is an odd multiple of 3, corollary 1.2.1 tells us that  $x+1$  divides  $F_b(x)$ , but  $x+1$  does not divide  $F_b(x+1)$ . So,

$$F_{n+1}(x) = x^{2^k-1} (x+1)^{2^k} g(x)^{2^k} \quad F_{n+1}(x+1) = x^{2^k} (x+1)^{2^k-1} g(x+1)^{2^k},$$

for some  $g(x) \in \mathbb{F}_2[x]$ , where  $g(x)$  and  $g(x+1)$  are both divisible by neither  $x$  nor  $x+1$ . So, we see that  $F_{n+1}(x)$  can be divided without remainder by  $x+1$  more times than  $f_{n+1}(x+1)$ , so  $\delta_n = 2$ . ■

This result was first proven by Boyles in [2] and resolves a conjecture of Sutner's [9].

We can repeatedly apply theorem 3.1 to obtain explicit formulas for  $d(n)$ .

**Corollary 3.1.1.**  *$n$  as  $n+1 = 2^s b$  for  $s \geq 0$  and  $b$  odd. Then*

$$d(n) = 2^s d(b-1) + \begin{cases} 2(2^s - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases}.$$

*Proof.* The result is a tautology for even  $n$  ( $s = 0$  and  $n+1 = b$ ), so we may continue assuming  $n$  is odd.

By induction on  $s$ . For  $s = 0$ ,

$$\begin{aligned}
n+1 &= 2^0 b \\
n &= b-1 \\
d(n) &= d(b-1) \\
&= 2^0 d(b-1) + \begin{cases} 2(2^0 - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases},
\end{aligned}$$

as desired. Assume that our claim holds for  $s \geq 0$ . Then for  $n + 1 = 2^{s+1}b$ ,

$$\begin{aligned}
d(n) &= d(2^{s+1}b - 1) \\
&= d(2(2^s b - 1) + 1) \\
&= 2d(2^s b - 1) + \begin{cases} 2 & 3 \mid 2^s b \\ 0 & \text{otherwise} \end{cases} \\
&= 2 \left( 2^s d(b - 1) + \begin{cases} 2(2^s - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases} \right) + \begin{cases} 2 & 3 \mid 2^s b \\ 0 & \text{otherwise} \end{cases} \\
&= 2^{s+1}d(b - 1) + \begin{cases} 2(2^{s+1} - 1) - 2 & 3 \mid b \\ 0 & \text{otherwise} \end{cases} + \begin{cases} 2 & 3 \mid b \\ 0 & \text{otherwise} \end{cases} \\
&= 2^{s+1}d(b - 1) + \begin{cases} 2(2^{s+1} - 1) & 3 \mid b \\ 0 & \text{otherwise,} \end{cases}
\end{aligned}$$

as desired. ■

We can also use theorem 3.1 to narrow down more when  $d(n) = 2$ .

**Corollary 3.1.2.** *If  $d(n) = 2$ , then  $n \equiv -1 \pmod{6}$ .*

*Proof.* Since 2 is not divisible by 4, corollary 1.4.1 tells us that  $n$  must be odd. So, we can write  $n = 2^k b - 1$  for  $b$  odd and  $k \geq 1$ . Using corollary 3.1.1, we can define

$$D_i = d(2^i b - 1) = \begin{cases} 2^i d(b - 1) + 2(2^i - 1) & 3 \mid b \\ 2^i d(b - 1) & \text{otherwise} \end{cases}.$$

If  $3 \nmid b$ , then we must have  $D_k = 2 = 2^k d(b - 1)$ . However, corollary 1.4.1 tells us that  $d(b - 1)$  is divisible by 4, which cannot be the case.

If  $3 \mid b$ , then we must have  $D_k = 2 = 2^k d(b - 1) + 2(2^k - 1)$ . If  $k \geq 2$ , then this expression is greater than 2, which cannot be the case. Thus, we must have  $k = 1$  and  $D_1 = 2 = 2d(b - 1) + 2$ , which simplifies to  $d(b - 1) = 0$ . Thus we may write  $b = 3j$ .

Solving for  $n$ , we find

$$\begin{aligned}
n &= 2^k b - 1 \\
&= 2b - 1 \\
&= 6j - 1,
\end{aligned}$$

so  $n \equiv -1 \pmod{6}$ , as desired. ■

We can now use this recurrence result to prove the impossibility of certain values of  $d(n)$ .

**Theorem 3.2.** *The smallest even  $k$  such that for all  $n \in \mathbb{N}$ ,  $d(n) \neq k$  is  $k = 26$ .*

*Proof.* Recall from lemma 1.4 that  $d(n)$  is always even. Through calculation, we can find that

$$\begin{array}{llllll}
d(5) = 2 & d(4) = 4 & d(11) = 6 & d(16) = 8 & d(29) = 10 & d(84) = 12 \\
d(23) = 14 & d(19) = 16 & d(101) = 18 & d(30) = 20 & d(59) = 22 & d(62) = 24.
\end{array}$$

We will show that  $d(n) = 26$  is impossible.

Assume for the sake of contradiction that there exists  $n \in \mathbb{N}$  such that  $d(n) = 26$ . Since 26 is not divisible by 4, corollary 1.4.1 that  $n$  must be odd. So, we may write  $n = 2m + 1$ . By theorem 3.1,

$$26 = d(n) = d(2m + 1) = 2d(m) + \begin{cases} 2 & 3 \mid m + 1 \\ 0 & \text{otherwise} \end{cases}.$$

Since lemma 1.4 tells us that  $d(m) = 13$  is not possible, we must have  $d(m) = 12$  and  $m + 1$  is divisible by 3. Let  $m + 1 = 2^s b$  for odd  $b$  and  $s \geq 0$ . By corollary 3.1.1,

$$d(m) = 2^s d(b - 1) + 2(2^s - 1) = 12.$$

The only possible integer solutions here are  $d(b - 1) = 12$  and  $s = 0$  or  $d(b - 1) = 5$  and  $s = 1$ . Since lemma 1.4 tells us that  $d(b - 1) = 5$  is impossible, the only remaining possibility is that  $d(m) = d(b - 1) = 12$  and  $s = 0$ . Thus,  $m$  must be even.

We know from lemma 1.1 that

$$d(m) = \deg(\gcd(F_{m+1}(x), F_{m+1}(x + 1))).$$

Since  $m + 1$  is odd lemma 1.3 tells us that there exists some square-free  $R_{m+1}(x)$  such that

$$F_{m+1}(x) = R_{m+1}(x)^2.$$

Define  $H_{m+1}(x)$

$$H_{m+1}(x) = \gcd(R_{m+1}(x), R_{m+1}(x + 1)) \quad \gcd(F_{m+1}(x), F_{m+1}(x + 1)) = H(x)^2.$$

Since  $R_{m+1}(x)$  is square-free, then  $H_{m+1}(x)$  is too. Since  $d(m) = 12$ ,  $\deg H_{m+1}(x) = 6$ . Notice too that  $H_{m+1}(x) = H_{m+1}(x + 1)$ , so our work in lemma 1.5 tells us that for  $y = x^2 + x$ , there exists square-free a cubic polynomial  $h_{m+1}(y) \in \mathbb{F}_2[y]$  such that

$$H_{m+1}(x) = h_{m+1}(y).$$

There are only three possible square-free cubic polynomials in  $\mathbb{F}_2[y]$  that are also 1 at  $y = 0$ :

1.  $C_1(y) = y^3 + 1$
2.  $C_2(y) = y^3 + y + 1$
3.  $C_3(y) = y^3 + y^2 + 1$ .

We will handle these case-by-case.

Define the “rank” of a polynomial  $P$  to be the smallest integer  $r$  such that  $P$  divides  $F_r$ . Calculating the rank of our three candidates for  $h_{m+1}(x)$ ,

$$\begin{aligned} \text{rank}((x^2 + x)^3 + 1) &= 85 \\ \text{rank}((x^2 + x)^3 + (x^2 + x) + 1) &= 63 \\ \text{rank}((x^2 + x)^3 + (x^2 + x)^2 + 1) &= 63 \end{aligned}$$

If either rank-63 polynomials  $C_2$  or  $C_3$  divides  $R_{m+1}(x)$ , then 63 divides  $m + 1$ . However, by definition of rank, both polynomials would divide  $R_{m+1}(x)$ .  $C_2$  and  $C_3$  are coprime and each of degree 6 in  $x$ , so  $\deg H_{m+1}(x) = \deg C_2(x) + \deg C_3(x) = 12$ , contradicting our assumption that the degree was 6. If  $C_1$  divides  $R_{m+1}$ , then 85 divides  $m + 1$ . Combining this with the fact that 3 divides  $m + 1$  we showed earlier, we see that 255 must divide  $m + 1$ .

Define

$$P(x) = x^4 + x^3 + 1 \quad Q(x) = x^4 + x^3 + x^2 + x + 1.$$

$P$  and  $Q$  are closed under translation ( $P(x+1) = Q(x)$ ), coprime to each other, and coprime with  $C_1$ . The ranks of  $P(x)$  and  $Q(x)$  are 15 and 17 respectively. Since both 15 and 17 divide 255, and 255 divides  $m+1$ , then both  $P(x)$  and  $Q(x)$  must divide  $H_{m+1}(x)$ . Therefore,

$$\deg H_{m+1}(x) \geq \deg C_1(x) + \deg P(x) + \deg Q(x) = 14,$$

which contradicts our assumption that the degree was 6.

Thus, there is no  $3 \mid m+1$  such that  $d(m) = 12$  and therefore no  $n$  such that  $d(n) = 26$ . ■

Now that we know that one impossible even value of  $d(n)$  exists, we can show an entire family of impossible even values also exists.

**Corollary 3.1.3.** *Suppose  $a \in \mathbb{N}$  is even and an impossible value of  $d(n)$ . Then  $2a+2$  is also an impossible value of  $d(n)$ .*

*Proof.* Suppose for the sake of contradiction that there exists  $n$  such that  $d(n) = 2a+2$ . Since  $2a+2$  is not divisible by 4, then corollary 1.4.1 tells us that  $n$  must be odd, so we may write  $n = 2m+1$ . So, by theorem 3.1,

$$2a+2 = d(n) = d(2m+1) = 2d(m) + \delta_m \quad \delta_m = \begin{cases} 2 & 3 \mid m+1 \\ 0 & \text{otherwise} \end{cases}.$$

If  $\delta_m = 0$ , then  $2d(m)$  is divisible by 4, which cannot be the case since we know the value is  $2a+2$ . If  $\delta_m = 2$ , then  $d(m) = a$ , contradicting the impossibility of  $a$  as an output of  $d$ . ■

Expanding the recurrence, if  $a$  is an impossible value, then  $2^t(a+2) - 2$  is also impossible for all  $t \geq 0$ . In other words, 26, 54, 110, 222, 446, ... are also all impossible values of  $d(n)$ .

## 4 Finite Fields

Although we have already been working in  $\mathbb{F}_2$ , we will now more directly use results from the study of finite fields.

**Lemma 4.1.** *Let  $p$  be an odd prime. Define  $h = \text{ord}_p(2)$  and  $c = v_p(2^h - 1)$ , where  $v_p(a)$  is the  $p$ -adic valuation of  $a$ . Then for every  $j \geq 1$ ,*

$$\text{ord}_{p^j}(2) = hp^{\max(0, j-c)}.$$

*Proof.* Suppose  $2^r \equiv 1 \pmod{p^j}$ . Then  $h$  divides  $r$ , so we may write  $r = ht$ . By the Lifting the Exponent Lemma,

$$\begin{aligned} v_p(2^r - 1) &= v_p(2^{ht} - 1) \\ &= v_p(2^h - 1) + v_p(t) \\ &= c + v_p(t). \end{aligned}$$

So,  $p^j \mid 2^{ht} - 1$  exactly when  $c + v_p(t) \geq j$ . The smallest possible  $t$  satisfying this inequality is  $t = p^{\max(0, j-c)}$ , so

$$\text{ord}_{p^j}(2) = hp^{\max(0, j-c)},$$

as desired. ■

**Definition 4.1.** *A prime number  $p$  is a Wieferich prime when*

$$p^2 \mid 2^{p-1} - 1.$$

Although infinitely any Wieferich primes are conjectured to exist, the only ones known to date are 1093 and 3511 [4], and any larger ones must be greater than  $2^{64}$  [3].

We will show an equivalent definition of Wieferich primes.

**Lemma 4.2.** *A prime number  $p$  is a Wieferich prime if and only if*

$$p^2 \mid 2^{\text{ord}_p(2)} - 1.$$

*Proof.* Let  $n = \text{ord}_p(2)$ . Then  $p - 1 = kn$  for some natural number  $k$ , and

$$2^{p-1} - 1 = 2^{kn} - 1.$$

So, by the Lifting the Exponent lemma,

$$v_p(2^{kn} - 1) = v_p(2^n - 1) + v_p(k).$$

However,  $k = (p - 1)/n$  is smaller than  $p$  and thus not divisible by  $p$ , so  $v_p(k) = 0$ . Thus,

$$v_p(2^{p-1} - 1) = v_p(2^n - 1).$$

So,  $p^2 \mid 2^{p-1} - 1$  if and only if  $p^2 \mid 2^{\text{ord}_p(2)} - 1$ , as desired. ■

**Corollary 4.2.1.** *Let  $p$  be an odd prime. If  $p$  isn't a Wieferich prime, then*

$$\text{ord}_{p^j}(2) = \text{ord}_p(2) p^{j-1}.$$

*Proof.* Lemma 4.1 tells us that

$$\text{ord}_{p^j}(2) = \text{ord}_p(2) p^{\max(0, j-c)}, \quad c = v_p(2^{\text{ord}_p(2)} - 1).$$

Thus it is sufficient to show that  $c = 1$ . By definition of  $\text{ord}_p(2)$ ,  $c \geq 1$ .  $p$  is not a Wieferich prime, so by lemma 4.2, for all  $t \geq 2$ ,

$$p^t \nmid 2^{\text{ord}_p(2)} - 1.$$

Thus  $c < 2$  and therefore must be exactly 1, as desired. ■

**Definition 4.2.** *Let  $b$  be coprime to  $n$ . Then “signed order” of  $b$  modulo  $n$  is*

$$\rho_b(n) = \min \{r > 0 : b^r \equiv \pm 1 \pmod{n}\}.$$

This is also sometimes called the “index” of  $\pm 1$  relative to  $b$ , written  $\text{ind}_b(\pm 1)$ .

**Lemma 4.3.** *Let  $p$  be an odd prime,  $j \geq 1$ , and  $H_j = \text{ord}_{p^j}(2)$ . Then*

$$\rho_2(p^j) = \begin{cases} H_j/2 & H_j \text{ is even} \\ H_j & H_j \text{ is odd} \end{cases}.$$

Moreover, when  $H_j$  is even,

$$2^{H_j/2} \equiv -1 \pmod{p^j}.$$

*Proof.* Assume  $H_j$  is even. Then by definition of  $H_j$ ,

$$\left(2^{H_j/2}\right)^2 \equiv 1 \pmod{p^j}.$$

Thus the only solutions are  $2^{H_j/2} \equiv \pm 1 \pmod{p^j}$ . However, 1 cannot be a solution; otherwise,  $H_j$  would not be minimal. So,

$$2^{H_j/2} \equiv -1 \pmod{p^j}.$$

Notice that if  $r \leq H_j/2$ , and  $2^r \equiv -1 \pmod{p^j}$ , then  $2^{2r} \equiv 1 \pmod{p^j}$ ; since  $H_j$  is minimal,  $H_j \leq 2r$ . Thus to satisfy both constraints we must have  $H_j/2 = r$ .

Now assume  $H_j$  is odd. Notice that by definition,

$$H_j = \text{ord}_{p^j}(2) = \min \{h > 0 : 2^h \equiv 1 \pmod{p^j}\},$$

so smaller solutions can only be of the form  $2^r \equiv -1 \pmod{p^j}$ . Then  $2^{2r} \equiv 1 \pmod{p^j}$ , which means  $H_j \mid 2r$ , and since  $H_j$  is odd,  $H_j \mid r$ . Thus  $H_j \leq r$ , so  $H_j$  is minimal, as desired. ■

We can use this notion of a signed order to find the extension degree of extensions of  $\mathbb{F}_2$ .

**Lemma 4.4.** *Let  $M > 1$  be odd; let  $\zeta$  have exact order  $M$ ; set  $\alpha = \zeta + \zeta^{-1}$ . Then*

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M).$$

*Proof.* A standard finite field fact says

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \min \{r > 0 : \alpha^{2^r} = \alpha\}.$$

By definition of  $\alpha$ ,  $\alpha^{2^r} = \zeta^{2^r} + \zeta^{-2^r}$ , so  $\alpha^{2^r} = \alpha$  is equivalent to  $\zeta^{2^r} + \zeta^{-2^r} = \zeta + \zeta^{-1}$ .

For any non-zero  $a, b \in \overline{\mathbb{F}_2}$ ,  $a + a^{-1} = b + b^{-1} \iff (a+b)(ab+1) = 0$ . So, either  $a = b$  or  $a = b^{-1}$ . Taking  $a = \zeta^{2^r}$  and  $b = \zeta$ , we obtain

$$\zeta^{2^r} = \zeta \quad \text{or} \quad \zeta^{2^r} = \zeta^{-1}.$$

Since  $\zeta$  has exact order  $M$ , we can equivalently write

$$2^r \equiv 1 \pmod{M} \quad \text{or} \quad 2^r \equiv -1 \pmod{M}.$$

So,  $\alpha^{2^r} = \alpha \iff 2^r \equiv \pm 1 \pmod{M}$ , and

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \min \{r > 0 : \alpha^{2^r} = \alpha\} = \rho_2(M),$$

as desired. ■

**Lemma 4.5.** *Let  $a$  be algebraic over  $\mathbb{F}_q$ . Then for every  $z \in \mathbb{F}_q(a)$ ,*

$$\text{Tr}_{\mathbb{F}_q(a)/\mathbb{F}_q}(z) = \sum_{r=0}^{[\mathbb{F}_q(a):\mathbb{F}_q]-1} z^{q^r}.$$

*Proof.* See [7]. ■

**Corollary 4.5.1.**

$$\text{Tr}_{\mathbb{F}_q(a)/\mathbb{F}_q}(1) = [\mathbb{F}_q(a) : \mathbb{F}_q].$$

*Proof.* Immediate from evaluating lemma 4.5 at  $z = 1$ . ■

**Lemma 4.6.** *Let  $M > 1$  be odd, and let  $\zeta$  have exact order  $M$ . Let  $\alpha = \zeta + \zeta^{-1}$ . Define*

$$G_M = \{\pm 2^r \pmod{M} : r \geq 0\} \subseteq (\mathbb{Z}/M\mathbb{Z})^\times.$$

*Then*

$$\text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u.$$

*Proof.* Lemma 4.4 showed that

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M).$$

So, by lemma 4.5,

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{r=0}^{\rho_2(M)-1} \zeta^{2^r} + \zeta^{-2^r}.$$

By the definition of  $\rho_2(M)$ , the residues  $2^0, -2^0, 2^1, -2^1, \dots, 2^{\rho_2(M)-1}, -2^{\rho_2(M)-1}$  are all distinct modulo  $M$ . Otherwise, we would obtain  $2^r \equiv \pm 2^s \pmod{M}$ , or equivalently,  $2^{r-s} \equiv \pm 1 \pmod{M}$  for some  $0 \leq s < r < d$ , which contradicts  $\rho_2(M)$  being minimal.

The residues are exactly the elements of  $G_M$ , so

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u,$$

as desired. ■

**Corollary 4.6.1.** *If  $G_M = (\mathbb{Z}/M\mathbb{Z})^\times$ , then*

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \mu(M) \pmod{2},$$

where  $\mu$  is the number-theoretic Möbius function.

*Proof.* From lemma 4.6, we know that

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u.$$

When  $G_M = (\mathbb{Z}/M\mathbb{Z})^\times$ , we also have

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{0 \leq u < M, \gcd(u, M)=1} \zeta^u = \mu(M) \pmod{2},$$

by Ramanujan's sum identity of roots of unity, as desired. ■

Now we may combine together many of our partial results to state a more significant result.

**Theorem 4.1.** *If  $p$  is not a Wieferich prime, then for all  $k \in \mathbb{N}$ ,*

$$d(p^k - 1) = d(p - 1).$$

*Proof.* Let  $p$  be a non-Wieferich prime. The case for  $p = 2$  was shown in [9], so we will consider the case when  $p$  is odd. For  $1 \leq j \leq k$ , let

$$R_j = \{\zeta + \zeta^{-1} : \mathrm{ord}(\zeta) = p^j\}.$$

By lemma 1.7,

$$\mathrm{roots} F_{p^k} = \bigcup_{j=1}^k R_j. \tag{1}$$

Furthermore, since  $p^k$  is odd, corollary 1.3.1 and lemma 1.3 tell us that  $F_{p^k}$  is the square of a square-free polynomial. In other words,  $F_{p^k}$  has  $(p^k - 1)/2$  distinct roots. Let  $h = \mathrm{ord}_p(2)$ . Since  $p$  is not Wieferich, corollary 4.2.1 tells us that

$$\mathrm{ord}_{p^j}(2) = \mathrm{ord}_p(2) p^{j-1},$$

and lemma 4.3 tells us that

$$\rho_2(p^j) = \rho_2(p) p^{j-1}.$$

So, by lemma 4.4, if  $\alpha \in R_j$ ,

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(p)p^{j-1}.$$

Suppose both  $\alpha$  and  $\alpha + 1$  are roots of  $F_{p^k}$  with  $\alpha \in R_j$  and  $\alpha + 1 \in R_\ell$ . Since  $\mathbb{F}_2(\alpha) = \mathbb{F}_2(\alpha + 1)$ , that is they are the same field with the same degree,  $\alpha$  and  $\alpha + 1$  must both be members of the same  $R_j$  and so  $j = \ell$ .

Suppose for the sake of contradiction that  $\alpha$  and  $\alpha + 1$  are both members of  $R_j$  for some  $j \geq 2$ . Set  $M = p^j$  and  $m = p^{j-1}$  and choose a primitive  $M$ th root  $\zeta$  such that  $\alpha = \zeta + \zeta^{-1}$ . Since  $\alpha + 1$  is also in  $R_j$ , there must exist  $s$  such that  $\gcd(s, p) = 1$  and  $\alpha + 1 = \zeta^s + \zeta^{-s}$ . Then

$$1 + \alpha + (\alpha + 1) = 1 + \zeta + \zeta^{-1} + \zeta^s + \zeta^{-s} = 0.$$

Equivalently,  $\zeta$  is a root of

$$P_s(x) = 1 + x + x^{M-1} + x^s + x^{M-s}.$$

Notice that  $\zeta^m$  is a primitive  $p$ th root. Let  $f(x)$  be the minimal polynomial of  $\zeta^m$  over  $\mathbb{F}_2$ . The degree of  $f(x)$  is  $\text{ord}_p(2)$ . Since  $f(\zeta^m) = 0$ ,  $\zeta$  is a root of  $f(x^m)$ . The degree of  $f(x^m)$  is  $\text{ord}_p(2)m = \text{ord}_{p^j}(2)$ . This is also the degree of the minimal polynomial of  $\zeta$  and so

$$f(x^m) \mid P_s(x).$$

So, every polynomial can be decomposed by exponents modulo  $m$ :

$$P_s(x) = \sum_{a=0}^{m-1} x^a P_a(x^m). \quad (2)$$

Since  $f(x^m) \mid P_s(x)$ , each  $P_a(y)$  must be divisible by  $f(y)$ . However, only exponent 0 among 0, 1,  $M-1$ ,  $s$ ,  $M-s$  is divisible by  $m$ , so  $P_0(y) = 1$ , a contradiction. Instead, we must conclude that no common roots of  $F_{p^k}(x)$  and  $F_{p^k}(x+1)$  are in  $R_j$  for  $j \geq 2$ .

So the only common roots of  $F_{p^k}(x)$  and  $F_{p^k}(x+1)$  must come from  $R_1$  and so we must conclude

$$d(p^k - 1) = d(p - 1).$$

■

We can combine this result with our recurrence when doubling  $n$  and adding 1.

**Corollary 4.6.2.** *Let  $p$  be an odd non-Wieferich prime. Then for all  $n, m \geq 0$ ,*

$$d(2^n p^m - 1) = \begin{cases} 0 & m = 0 \\ 2^n d(p - 1) & m \geq 1, p \neq 3 \\ 2^{n+1} - 2 & m \geq 1, p = 3 \end{cases}.$$

*Proof.* The  $m = 0$  case reduces to what Sutner already showed [9]. For the other cases, we first apply corollary 3.1.1 to obtain

$$d(2^n p^m - 1) = \begin{cases} 0 & m = 0 \\ 2^n d(p^m - 1) & m \geq 1, p \neq 3 \\ 2^{n+1} - 2 & m \geq 1, p = 3 \end{cases}.$$

We then apply theorem 4.1 to change  $d(p^m - 1)$  to  $d(p - 1)$ .

■

**Corollary 4.6.3.** *For all  $n, m \geq 1$ , the following relationships hold*

$$1. \ d(3^n - 1) = 0$$



2.  $d(5^n - 1) = 4$
3.  $d(7^n - 1) = 0$
4.  $d(17^n - 1) = 8$
5.  $d(31^n - 1) = 20$
6.  $d(127^n - 1) = 56$
7.  $d(257^n - 1) = 144$
8.  $d(683^n - 1) = 132$
9.  $d(8191^n - 1) = 4004$
10.  $d(2^n 3^m - 1) = 2^{n+1} - 2$
11.  $d(2^n 5^m - 1) = 2^{n+2}$
12.  $d(2^n 7^m - 1) = 0$
13.  $d(2^n 17^m - 1) = 2^{n+3}$
14.  $d(2^n 31^m - 1) = 2^{n+2} 5$
15.  $d(2^n 127^m - 1) = 2^{n+3} 7$
16.  $d(2^n 257^m - 1) = 2^{n+4} 9$
17.  $d(2^n 683^m - 1) = 2^{n+2} 33$
18.  $d(2^n 8191^m - 1) = 2^{n+2} 1001$

You may have noticed that for many sizes, we have  $d(p-1) = 0$ . We can actually classify some  $p$  for which this is the case.

**Theorem 4.2.** *Let  $p$  be a prime one less than a multiple of 4 where*

$$\{\pm 2^r \pmod{p} : r \geq 0\} = (\mathbb{Z}/p\mathbb{Z})^\times.$$

*Then  $d(p-1) = 0$ .*

*Proof.* Assume for the sake of contradiction that  $d(p-1) \neq 0$ . Then  $F_p(x)$  and  $F_p(x+1)$  have a common factor and thus a common root  $\alpha$ , so

$$F_p(\alpha) = F_p(\alpha+1) = 0.$$

By lemma 1.7, there must exist non-trivial  $p$ th roots of unity  $\zeta$  and  $\eta$  such that

$$\alpha = \zeta + \zeta^{-1} \quad \alpha + 1 = \eta + \eta^{-1}.$$

By lemmas 4.4 and 4.3,

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = [\mathbb{F}_2(\alpha+1) : \mathbb{F}_2] = \rho_2(p) = \frac{p-1}{2}.$$

Since  $p \equiv 3 \pmod{4}$ , this value is odd. Applying the linearity of the trace and corollary 4.5.1,

$$\text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha+1) = \text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) + \text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(1) = \text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) + 1 = 0.$$

However, we can see that  $\mathbb{F}_2(\alpha+1) = \mathbb{F}_2(\alpha)$  ( $\alpha$  and  $\alpha+1$  can be obtained from each other by adding 1), and so by corollary 4.6.1,

$$\text{Tr}_{\mathbb{F}_2(\alpha+1)/\mathbb{F}_2}(\alpha+1) = \text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \mu(p) \pmod{2} = 1.$$

Thus we have reached a contradiction. ■

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